

Semantic Theory 2026: Exercise 7 Key

Question 1

Recall the syntax for well-formed formulas (WFFs) in Dynamic Predicate Logic (DPL):

- All atomic formulas ($R(t_1, \dots, t_n)$ for $R \in PRED^n$, $t_1, \dots, t_n \in TERM$) are WFFs
- If $x \in VAR$, then $\exists x$ is a WFF
- If ϕ and ψ are WFFs, then $\sim\phi$ and $(\phi \cdot \psi)$ are WFFs
- Nothing else is a WFF

Translate the following natural language utterances into DPL. You may treat underlined expressions as single terms (e.g. “came to” $\Rightarrow$ $come\text{-}to(\dots)$):

- a. “If John runs, he will pull a muscle.”
 $run(j') \rightarrow (\exists x \cdot muscle(x) \cdot pull(j', x))$
 $= \sim(run(j') \cdot \sim(\exists x \cdot muscle(x) \cdot pull(j', x)))$
- b. “There was a party. A man came to the party. He was hungry.”
 $(\exists x \cdot party(x)) \cdot (\exists y \cdot man(y) \cdot come\text{-}to(y, x)) \cdot hungry(y)$
 $= \exists x \cdot party(x) \cdot \exists y \cdot man(y) \cdot come\text{-}to(y, x) \cdot hungry(y)$
- c. “There is a farmer. She owns a donkey. If the donkey is hungry, she feeds it.”
 $(\exists x \cdot farmer(x)) \cdot (\exists y \cdot donkey(y) \cdot own(x, y)) \cdot (hungry(y) \rightarrow feed(x, y))$
 $= \exists x \cdot farmer(x) \cdot \exists y \cdot donkey(y) \cdot own(x, y) \cdot \sim(hungry(y) \cdot \sim feed(x, y))$

Question 2

Translate the DPL formulas ϕ you constructed for 1a-c to FOL formulas $\phi^\circ = \langle \phi \rangle \top$, using the rules introduced in the lecture.

- i. $\langle \perp \rangle \psi = \perp$
- ii. $\langle \top \rangle \psi = \psi$
- iii. $\langle P(x_1, \dots, x_n) \rangle \psi = P(x_1, \dots, x_n) \wedge \psi$
- iv. $\langle \exists x \rangle \psi = \exists x[\psi]$
- v. $\langle \phi_1 \cdot \phi_2 \rangle \psi = \langle \phi_1 \rangle (\langle \phi_2 \rangle \psi)$
- vi. $\langle \sim\phi \rangle \psi = \neg(\langle \phi \rangle \top) \wedge \psi$

Show each step of the derivation, and indicate the rule applied ((i)-(vi)) at each step:

a.

$$\begin{aligned}
& (\sim(\text{run}(j') \cdot \sim(\exists x \cdot \text{muscle}(x) \cdot \text{pull}(j', x))))^\circ \\
&= \langle \sim(\text{run}(j') \cdot \sim(\exists x \cdot \text{muscle}(x) \cdot \text{pull}(j', x))) \rangle \top \\
&= \neg(\langle \text{run}(j') \cdot \sim(\exists x \cdot \text{muscle}(x) \cdot \text{pull}(j', x)) \rangle \top) \wedge \top \quad (\text{vi}) \\
&= \neg(\langle \text{run}(j') \rangle (\langle \sim(\exists x \cdot \text{muscle}(x) \cdot \text{pull}(j', x)) \rangle \top)) \wedge \top \quad (\text{v}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x \cdot \text{muscle}(x) \cdot \text{pull}(j', x) \rangle \top) \wedge \top)) \wedge \top \quad (\text{v}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x \rangle (\langle \text{muscle}(x) \cdot \text{pull}(j', x) \rangle \top) \wedge \top)) \wedge \top \quad (\text{v}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x \rangle (\langle \text{muscle}(x) \rangle (\langle \text{pull}(j', x) \rangle \top)) \wedge \top)) \wedge \top \quad (\text{v}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x \rangle (\langle \text{muscle}(x) \rangle (\text{pull}(j', x) \wedge \top)) \wedge \top)) \wedge \top \quad (\text{iii}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x \rangle (\text{muscle}(x) \wedge \text{pull}(j', x) \wedge \top)) \wedge \top)) \wedge \top \quad (\text{iii}) \\
&= \neg(\langle \text{run}(j') \rangle (\neg(\langle \exists x [\text{muscle}(x) \wedge \text{pull}(j', x) \wedge \top] \rangle \wedge \top)) \wedge \top \quad (\text{iv}) \\
&= \neg(\text{run}(j') \wedge \neg(\langle \exists x [\text{muscle}(x) \wedge \text{pull}(j', x) \wedge \top] \rangle \wedge \top)) \wedge \top \quad (\text{iv}) \\
&= \neg(\text{run}(j') \wedge \neg \exists x [\text{muscle}(x) \wedge \text{pull}(j', x)]) \\
&= \text{run}(j') \rightarrow \exists x [\text{muscle}(x) \wedge \text{pull}(j', x)]
\end{aligned}$$

b.

$$\begin{aligned}
& (\exists x \cdot \text{party}(x) \cdot \exists y \cdot \text{man}(y) \cdot \text{come-to}(y, x) \cdot \text{hungry}(y))^\circ \\
&= \langle \exists x \cdot \text{party}(x) \cdot \exists y \cdot \text{man}(y) \cdot \text{come-to}(y, x) \cdot \text{hungry}(y) \rangle \top \\
&= \langle \exists x \rangle (\langle \text{party}(x) \cdot \exists y \cdot \text{man}(y) \cdot \text{come-to}(y, x) \cdot \text{hungry}(y) \rangle \top) \quad (\text{v}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \cdot \text{man}(y) \cdot \text{come-to}(y, x) \cdot \text{hungry}(y) \rangle \top)) \quad (\text{v}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \rangle (\langle \text{man}(y) \cdot \text{come-to}(y, x) \cdot \text{hungry}(y) \rangle \top))) \quad (\text{v}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \rangle (\langle \text{man}(y) \rangle (\langle \text{come-to}(y, x) \cdot \text{hungry}(y) \rangle \top)))) \quad (\text{v}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \rangle (\langle \text{man}(y) \rangle (\langle \text{come-to}(y, x) \rangle (\text{hungry}(y) \wedge \top)))))) \quad (\text{iii}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \rangle (\langle \text{man}(y) \rangle (\text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top)))) \quad (\text{iii}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y \rangle (\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top))) \quad (\text{iii}) \\
&= \langle \exists x \rangle (\langle \text{party}(x) \rangle (\langle \exists y [\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top] \rangle)) \quad (\text{iv}) \\
&= \langle \exists x \rangle (\text{party}(x) \wedge \langle \exists y [\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top] \rangle) \quad (\text{iii}) \\
&= \langle \exists x \rangle (\text{party}(x) \wedge \langle \exists y [\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top] \rangle) \quad (\text{iv}) \\
&= \exists x [\text{party}(x) \wedge \langle \exists y [\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y) \wedge \top] \rangle] \quad (\text{iv}) \\
&= \exists x [\text{party}(x) \wedge \exists y [\text{man}(y) \wedge \text{come-to}(y, x) \wedge \text{hungry}(y)]]
\end{aligned}$$

c.

$$\begin{aligned}
& (\exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y)))^\circ \\
&= \langle \exists x \cdot F(x) \cdot \exists y \cdot D(y) \cdot O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y)) \rangle \top \\
&= \langle \exists x \rangle (F(x) \cdot \exists y \cdot D(y) \cdot O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y))) \top \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\exists y \cdot D(y) \cdot O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y))) \top) \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (D(y) \cdot O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y))) \top))) \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (O(x, y) \cdot \sim(H(y) \cdot \sim feed(x, y))) \top)))) \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\langle \sim(H(y) \cdot \sim feed(x, y)) \rangle \top)))))) \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\neg(\langle H(y) \cdot \sim feed(x, y) \rangle \top) \wedge \top)))))) \quad (vi) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\neg(\langle H(y) \rangle (\langle \sim feed(x, y) \rangle \top)) \wedge \top)))))) \quad (v) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\neg(\langle H(y) \rangle (\neg(\langle feed(x, y) \rangle \top) \wedge \top)) \wedge \top)))))) \quad (vi) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\neg(\langle H(y) \rangle (\neg(feed(x, y) \wedge \top) \wedge \top)) \wedge \top)))))) \quad (iii) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (\langle O(x, y) \rangle (\neg(H(y) \wedge \neg(feed(x, y) \wedge \top) \wedge \top) \wedge \top)))))) \quad (iii) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (\langle D(y) \rangle (O(x, y) \wedge \neg(H(y) \wedge \neg(feed(x, y) \wedge \top) \wedge \top) \wedge \top)))) \quad (iii) \\
&= \langle \exists x \rangle (\langle F(x) \rangle (\langle \exists y \rangle (D(y) \wedge O(x, y) \wedge \neg(H(y) \wedge \neg(feed(x, y) \wedge \top) \wedge \top) \wedge \top))) \quad (iii) \\
&= \langle \exists x \rangle (F(x) \wedge \exists y [D(y) \wedge O(x, y) \wedge \neg(H(y) \wedge \neg(feed(x, y) \wedge \top) \wedge \top) \wedge \top]) \quad (iv) \\
&= \exists x [F(x) \wedge \exists y [D(y) \wedge O(x, y) \wedge \neg(H(y) \wedge \neg(feed(x, y) \wedge \top) \wedge \top) \wedge \top]] \quad (iv) \\
&= \exists x [F(x) \wedge \exists y [D(y) \wedge O(x, y) \wedge \neg(H(y) \wedge \neg feed(x, y))]] \\
&= \exists x [F(x) \wedge \exists y [D(y) \wedge O(x, y) \wedge (H(y) \rightarrow feed(x, y))]]
\end{aligned}$$